

Technical Document #2: Retrieval Augmented Generation - Comprehensive Overview

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Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Query decomposition breaks complex queries into simpler sub-queries.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Hybrid search combines keyword-based and semantic search.